

GIZMO Operations Guide

Never lift the safety ground. Follow SOP004 and the current work plan.

Installed system

GIZMO is already deployed. SOP004 authorizes its installation and use. The initial grounding setup will use one saturable inductor.

Item	Recorded setting
Excitation	1436 Hz
Alarm threshold	100 Ω
Runtime / OPC UA model	0.5.1 / 1.4.0
Monitoring server	dune-fd-test01
History available	Since July 27, 2026

These are stored settings, not live confirmation. As of September 30, 2026, the instrument is unreachable from the monitoring server. The latest retained sample is September 13, 2026, at 07:11:34 UTC. The server dashboard and historian are running.

Connections after installation or maintenance

With the rack de-energized, check each cable against its installed label and the approved drawing:

- Pearson CT output to GIZMO CT input; the CT observes the intended safety-ground path.
- Stimulus center to the designated Detector Ground point; return to the designated rack/infrastructure reference.
- Beacon and audible output to their labeled connectors.
- Network to the assigned controls port; approved DC supply to the chassis.
- One saturable inductor in the initial setup, with its protection and safety-ground connections verified by the electrical lead.

Changes to grounding, CT routing, stimulus, or calibration require the applicable work controls. Do not improvise a bypass or move connections to clear an alarm.

Start of shift

1. Open the GIZMO dashboard. Check the sample time and connection status before reading the alarm or impedance.
2. Confirm that the displayed frequency and threshold match the recorded settings. Report any unexplained change.
3. Check alarm, measurement quality, and recent history. A clear alarm is usable only with Good status and a fresh sample.
4. Record the check and any active welding or temporary bond in Procore Daily Log > GIZMO. Hand over unresolved alarms and outages.

HIGH Z is a range indication, not a resistance value. Do not replace it with a number. The last stored sample had Uncertain quality, an uncertain clear alarm, and a latched alarm; it does not establish the present ground condition.

If an alarm appears

1. Check whether an authorized temporary bond is active.

2. If the alarm is unexpected, stop new ground-affecting work. Record the time and alarm reason; notify the activity coordinator and grounding/GIZMO contact.
3. Check what changed: welding returns, temporary leads, cable shields, piping, scaffolding, or tools. Investigate under the work plan.
4. Keep the data and record the disposition. Do not change the threshold to make the alarm disappear.

If data stops

Treat stale or non-Good data as unavailable monitoring. A running dashboard or an old clear alarm is not proof that GIZMO is working.

1. Record the last sample time and notify the GIZMO and SC/DPS contacts.
2. Pause ground-affecting work unless the work plan permits an approved alternate monitor. Unrelated work need not stop solely because the data is stale.
3. Check the instrument power and network through the responsible on-site contact. The server reports loss of reachability; the cause has not been established.
4. After connection returns, verify advancing timestamps, usable measurement/alarm quality, and resumed history before relying on GIZMO again.

Welding or a planned bond

1. Before work, confirm the work package and return/bond arrangement. Record the starting condition and actual bond time.
2. Keep acquisition running. A planned bond can produce an alarm and mask an additional accidental short.
3. Inhibit annunciation only through the approved method; retain raw alarms and record the inhibition.
4. After work, remove temporary leads/bonds as specified, record the removal time, and confirm the normal condition returns. Close the window only after the responsible leads accept the result.

Server check

On dune-fd-test01, run:

```
/storage/gizmo-monitor/bin/status
```

It reports the historian, dashboard, upstream connection, stored samples, and disk space. Check upstream connectivity and the newest GIZMO sample, not just whether the processes are running.

For a local dashboard connection:

```
ssh -N -L 127.0.0.1:18081:127.0.0.1:18081 dune-fd-test01
```

Then open <http://127.0.0.1:18081/> in the browser. Use the configured SSH access for the server.

Shutdown

Coordinate any loss of monitoring before shutdown. Use SOP004 and the deployed operating procedure to disable excitation and power down. Leave the permanent safety-ground path intact. Record the time and reason.